Actual Rjoint network: asymmetric inputs, independent parameters

ACTOR (training and deployment)

Candidate variables
$C_t \in \mathbb{R}^{N_t \times 25}$

Node 8 + MIP 53
$x_t \in \mathbb{R}^{61}$

LayerNorm + Linear
25 → 256

LayerNorm + Linear
61 → 256

Broadcast tree + concatenate
Linear: 512 → 256

Transformer encoder
5 layers, 8 heads, d=256, FFN=1024

tree context

BiMatching: tree ↔ candidates
$N_t \times 256$ refined features

TreeGate head, one logit / candidate
256 → 128 → ... → 2 → 1

Mask padding + Softmax
$\pi_\theta(a_t \mid o_t)$

Categorical sampling
SCIP branches on selected variable

CRITIC (training only)

Candidate variables
$C_t \in \mathbb{R}^{N_t \times 25}$

Node 8 + MIP 53
$x_t \in \mathbb{R}^{61}$

LayerNorm + Linear
25 → 256

LayerNorm + Linear
61 → 256

Transformer encoder
5 layers, 8 heads, d=256

BiMatching: tree ↔ candidates
$N_t \times 256$ refined features

Masked mean over candidates + tree
Aggregation MLP: 512 → 256 → 256

Frontier input: 69
8 x 8 histogram
+ 5 scalars
MLP 69 → 64 → 32
Projection 32 → 256

Add training-only context
$z_{base} + W_{time}\tau_t + W_{extra}f(g_t)$

Remaining time
$\tau_t : 1 \to 256$

TreeGate value head
256 → 128 → ... → 2 → 1

Raw-unit value
$V_\phi = \mu + \sigma z$

Actor and Critic do NOT share weights. Orange modules are absent from Actor inference.

R0 removes the frontier branch; B0 also removes the time branch and static value calibration. No recurrent/Mamba module is used.